Rule-Based Classification of Edible and Poisonous Mushrooms - DISCUS

1. Introduction

In this report, we examine the challenge of telling apart edible and poisonous mushrooms using a data-driven method. Correctly identifying wild mushrooms matters for both ecology and public health. However, many edible and toxic species look alike, posing a real risk for collectors. Many poisonous mushrooms closely resemble safe ones, which makes it tough for non-experts to tell them apart. Therefore, it is essential to have reliable and easy methods for identification.

The National Audubon Society has a long mission to inspire people to explore, understand, and protect nature. For more than forty years, the Society has published reliable Field Guides that are both scientifically sound and easy for the public to use. To meet changing user needs and new technology, the Society plans to turn its Mushroom Field Guide into a mobile app that helps with real-time decisions in the field.

The Data-Intensive Science Centre (DISCUS) at the University of Sussex has been asked to help with this project. I have been invited to contribute to this effort as a data scientist at the Data-Intensive Science Centre (DISCUS) at the University of Sussex to develop a data-driven rule set that could help a non-expert distinguish between edible and toxic mushrooms. As a data scientist at DISCUS, I need to develop a rule set that could be integrated with a guided checklist within the app designed by the Society.

The main goal of the report is to examine the chosen dataset of mushroom observations we received. I will be using data science methods to establish a clear set of rules for the proposed application. To reach this goal, I look at a well-known mushroom dataset with physical characteristics and edibility labels. I use exploratory data analysis, statistical hypothesis testing, and a rule extraction method based on decision tree models. The rules I create are meant to act as a checklist for decision-making. This supports safer foraging practices while recognizing the uncertainties involved.

2. Data

2.1 Data Source

The dataset used in the analysis was obtained from the National Audubon Society and originated from a common mushroom classification dataset. The dataset contains descriptions of sample mushrooms for 23 species of gilled mushrooms in both the Agaricus and Lepiota genera. The two families of mushrooms include known edible species, in addition to highly poisonous ones. The dataset used in the project was chosen for the common nature of the mushrooms it describes. Each species in the dataset has been classified by experts as:

- Definitely edible
- Definitely poisonous

Because the documentation for the dataset followed best practices in public health for the categorization of edibility for the categories of mushrooms that were not known or poisonous, the category of mushrooms that were not known for their edibility included the category of poisonous mushrooms.

2.2 Dataset Structure

The analysis is based on two text files provided by the Society

1. Data File (agaricus-lepiota-data.txt)

This file contains observational records of mushrooms, containing:

- 8,124 mushroom samples
- 23 features, of which one is the class attribute

2. Attribute Information File (agaricus-lepiota-info.txt)

This file provides data describing:

- The meaning of each feature.
- A set of features describing physical characteristics such as cap shape, cap color, odor, gill size, stalk properties, habitat, and spore print color.
- The coded values used for each attribute.
- Background information about mushroom species included in the dataset.

2.3 Target Variable

The class labels are converted to lowercase strings and encoded with descriptive labels.

i.e. **Edible' for 'e' and 'Poisonous' for 'p.'**

Exploratory data analysis gives that the sample is well represented and that there is a good mix of edible and poisonous mushrooms. This is an important factor because it ensures that the data is not skewed to a certain classification.

2.4 Feature Description

Each mushroom specimen is described using **22 categorical features**, all of which correspond to observable characteristics that a person in the field could, in principle, identify. These features fall into several broad categories:

- **Cap features**: shape, surface texture, and colour
- **Physical reactions**: bruising behaviour when handled
- **Sensory properties**: odour
- **Gill features**: attachment, spacing, size, and colour
- **Stalk features**: shape, root type, surface texture, and colour above and below the ring
- **Veil and ring characteristics**: veil type and colour, ring number and ring type
- **Reproductive indicators**: spore print colour
- **Ecological context**: population density and habitat

2.5 Data Limitations

- The dataset consists of hypothetical mushroom samples, not real ones.
- Some characteristics have low variance values, which can decrease their level of importance for data analysis
- There were missing values in the "stalk-root" attribute.

2.6 Data Cleaning and Wrangling

Several data preparation steps were required before analysis:

1. **Column Naming**
   Since there were no headers in the raw data file, names of the features were assigned using the descriptions of the attributes in the metadata file. This was done by assigning a name to each column based on the attribute.

2. **Categorical decoding**
   All feature columns were encoded using the OrdinalEncoder, which encodes categorical values as integer ordinal. That would mean the unknown categories are converted to 1 to handle unexpected data.
3. **Handling missing values**
   Missing values in some of the features are represented as ‘?’. These records were not removed; instead, missing values were kept and treated as an explicit “unknown” category. This will maintain information and reflect realistic conditions in a field where not all characteristics may be observable. The stalk-root feature had 2,480 missing values, which was a significant number.
4. **Low-variance feature removal**
   The veil-type remained more or less constant in the observations and had almost the same value for all the mushrooms. Since a chi-square test requires variability in the data, this variable is not considered for the test.

Data cleaning led to improved consistency in the dataset by filling missing values with python NaN and then encoding all features in the dataset, which were found to be of type categorical. Removing an attribute with missing values is not a good way of handling missing data in mushroom data classification, since every single attribute has a possible role in whether a mushroom is safe for consumption or not.

3. Methods

3.1 Exploratory Data Analysis (EDA)

Initial exploratory analysis focused on:

- **Dataset size and structure**
  - Total Mushroom Samples: 8,124
  - Total Features: 22 categorical attributes plus 1 target class variable
- **Distribution of the target variable**
  - Edible (e): 4,208 samples (51.8%)
  - Poisonous (p): 3,916 samples (48.2%)
- **Feature cardinality (number of unique values per feature and missing values)**
  - stalk-root: 2,480 missing values
  - Unique values as per feature are shown in **Table 1**

3.2 Correlation Analysis

Although the dataset was categorical, the ordinal encodings are used in finding the correlations of the data features and the target class. These correlations may not lead to cause. However, they are a starting point for the ranking of the data features in the ordering of classes. The veil type characteristic has nearly no correlation with the edibility class, implying that it does not add anything significant to the model as a source of discrimination. Therefore, this feature was excluded during preprocessing.

3.3 Chi-Square Hypothesis Testing

Chi-Square test of independence was carried out to determines whether there is a significant variation in the distribution of values for the attribute between mushrooms that are edible and that are poisonous.

Hypotheses:

- Null hypothesis (H_0): The feature is independent of mushroom edibility.
- Alternative hypothesis (H_1): The feature is associated with mushroom edibility.

For each categorical feature:

- A contingency table was constructed against the class label.
- The chi-square statistic and corresponding p-value were computed.

Features with large chi-square statistics and very small p-values were interpreted as having strong evidence against the null hypothesis.

The test produces a **p-value**, which is compared against a predefined significance level of 5% i.e. **α = 0.05**.

The decision rules applied in this study are as follows:

- If p-value < 0.05
  - o Reject the null hypothesis (H_0)
  - o The feature is **statistically significant** and shows evidence of association with mushroom edibility.
- If p-value ≥ 0.05
  - o Fail to reject the null hypothesis (H_0)
  - o The feature is **not statistically significant**, and there is insufficient evidence to conclude an association with mushroom edibility.

These rules were used equally for each feature, and those features with highest importance were chosen for model building using a rule-based model approach.

3.4 Decision Tree Classifier

After the hypothesis test, a Decision Tree classifier was applied to model the relationship between the properties of mushrooms and their edibility. The Decision Tree was based on the interpretability of the results as well as the ability of the algorithm to provide simple rules for the classification.

The model parameters for avoiding overfitting were introduced, such as maximum tree depth and minimum samples per leaf node. The model is therefore forced to learn meaningful patterns rather than patterns in noises.

3.5 Rule Extraction Procedure

Once trained, the decision tree was traversed to extract:

- The conditions leading to each leaf
- The predicted class at that leaf
- The support (number of observations covered)
- The confidence (proportion of observations matching the predicted class)

After training the Decision Tree model, the tree is traversed from the root node downward. Each non-leaf node corresponds to a given conditional split based on a mushroom attribute, while each branch corresponds to an attribute value. traversal of the tree from leaf to root constitutes an if-then rule.

“If (set of observable conditions), then mushroom is edible/poisonous with X% confidence.”

These rules were compiled into a structured rulebook suitable for translation into app logic.

The rule extraction approach is organized in the following way:

- **Identification of the Root Node**
  The root node is termed as the attribute with the greatest information gain. In this analysis, the root node will always be referring to the odor attribute, symbolizing its significance in classifying mushrooms.

- **Path**
  Beginning from root, traversal is done along all branches in the tree. At each internal node, the splitting condition is noted as part of the rule.
- **Leaf Node Assignment**
  Every path ends at a leaf node, including the label of the predicted class (whether the item is edible or poisonous) as the conclusion of the rule.
- **Rule Creation**
  These recorded conditions are then joined together in a sequential fashion in order to create a full if-then statement.

4. Results

4.1 Feature cardinality and Correlation Analyses

This table shows the features cardinality analysis with columns for unique values, missing data counts and Correlation with class.

Feature	Unique	Missing	Correlation with Class
gill-size	2	0	0.540024357
population	6	0	0.298685536
habitat	7	0	0.217179278
cap-surface	4	0	0.178446126
spore-print-color	9	0	0.171960974
stalk-root	4	2480	0.167975282
veil-color	4	0	0.145141592
gill-attachment	2	0	0.12919986
cap-shape	6	0	0.052950564
cap-color	10	0	-0.031384087
odor	9	0	-0.093551644
stalk-shape	2	0	-0.102019017
stalk-color-below-ring	9	0	-0.146730283
stalk-color-above-ring	9	0	-0.154002725
ring-number	3	0	-0.214366472
stalk-surface-below-ring	4	0	-0.298800552
stalk-surface-above-ring	4	0	-0.334592749
gill-spacing	2	0	-0.348386785
ring-type	5	0	-0.411771386
bruises?	2	0	-0.501530377
gill-color	12	0	-0.530566191
veil-type	1	0	Nan

Table 1: Feature Cardinality and Correlation Summary

In table 1 above, we observe the following some top positive correlations with the target features most strongly associated with edible mushrooms:

- gill-size ($\rho = 0.540$)
- population ($\rho = 0.299$)

also, some top negative correlation with the target features strongest associated with poisonous mushrooms:

- gill-color ($\rho = -0.530$)
- bruises? ($\rho = -0.501$)

4.2 Exploratory Findings

The dataset consists of approximately several thousand examples of mushrooms, with approximately an equal number of both edible and poisonous examples. It was not seen that the class imbalance was severe.

Feature analysis revealed:

- Main features have low to moderate cardinality
- Missing data are present but sparse
- Certain features like odor and spore print color, strongly correlate with edibility

4.3 Chi-Square Test Results

Feature	Chi-Square Statistic	p-value
odor	7659.72674	0
spore-print-color	4602.03317	0
gill-color	3765.714086	0
ring-type	2956.619278	0
stalk-surface-above-ring	2808.286287	0
stalk-surface-below-ring	2684.474076	0
gill-size	2366.834257	0
stalk-color-above-ring	2237.898496	0
stalk-color-below-ring	2152.390891	0
bruises?	2041.415647	0
population	1929.740891	0
habitat	1573.777261	0
gill-spacing	984.143333	5.023E-216
stalk-root	638.2637627	5.104E-138
cap-shape	489.9199536	1.196E-103
cap-color	387.597769	6.0558E-78
ring-number	374.7368308	4.2358E-82
cap-surface	315.0428312	5.5184E-68
veil-color	191.2237015	3.321E-41
gill-attachment	133.9861813	5.5017E-31
stalk-shape	84.14203827	4.6047E-20

Table 2: Chi-Square Test Feature Summary

In table 2 above, from this we can see that for each feature, the p-value is much less than 0.05 (in fact, most of them are virtually 0), so H_0 is rejected for all. This implies:

- Every attribute is informative with respect to distinguishing edible and poisonous mushrooms.
- The probability of observing such extreme chi-square statistics if there were no relationship is essentially zero.

So statistically at the 5% percent level, no feature in the table is insignificant to the class label.

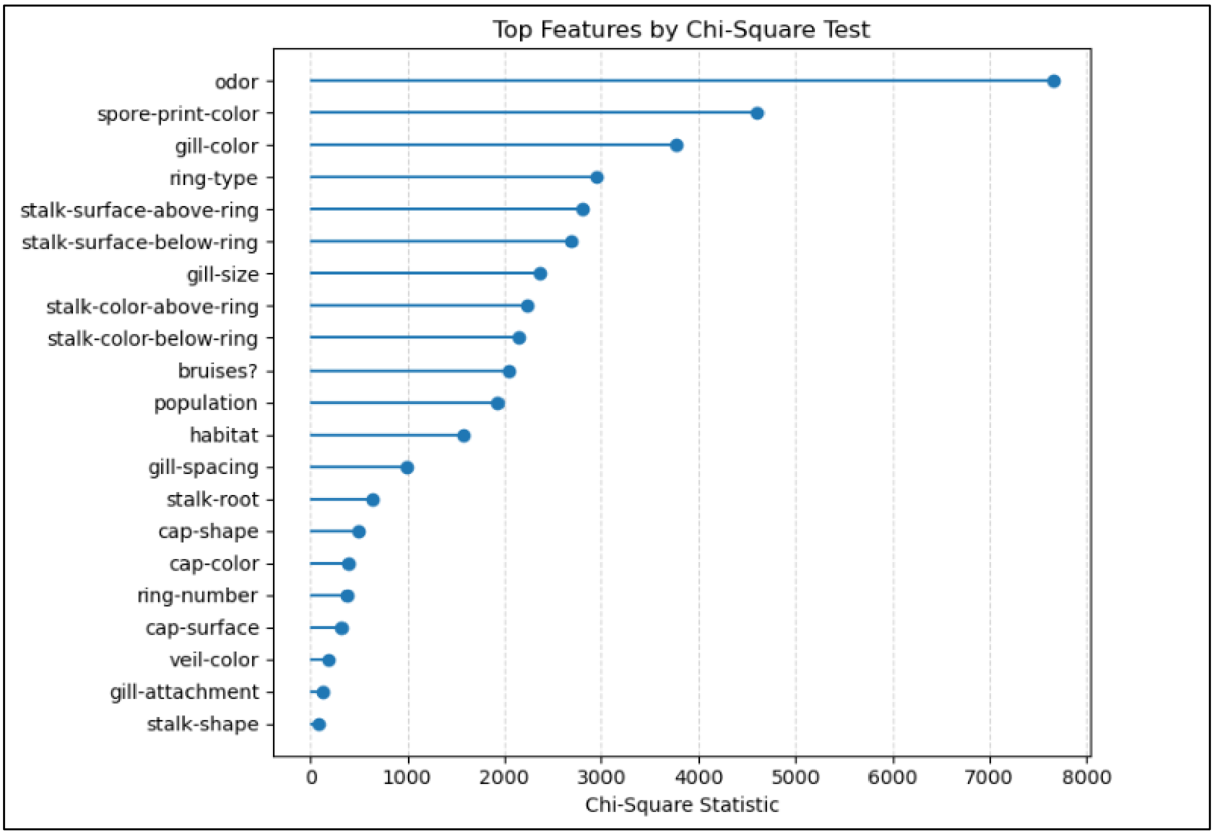

Figure 1: Chi-Square Statistic for each class

The Chi-square test has been able to point out certain characteristics which have a very high degree of statistical association with mushroom toxicity. Notably, **Odor** has been shown to have the highest degree of Chi-square Statistic, which means certain odors have a very predictive role in identifying poisonous mushrooms. Therefore, we also witness the importance in Chi-square Statistic, the greater the value, the stronger the evidence for dependence.

4.4 Decision Tree and Rule Set

Ultimately, the set of rules arrived at through the decision tree model is comprised of 15 unique classification statements that correspond to a particular set of mushroom attributes with the potential for being classified as either edible or poisonous. Although the ability to arrive at such a simple set of statements is certainly an effective element of the approach that was taken, an analysis of the statements made suggests both strengths as well as limitations.

Each rule is associated with a distinct root-to-leaf path in a decision tree. Along this root-to-leaf path, a split in the data is iterated using categorical values of input features that maximize class separation at each iteration. Once the trained tree was obtained, each of its leaf nodes was converted to a specific if-then rule. For each of these rules, two important data-driven measures were calculated: support, which represented the number of data points that matched a rule, and confidence, which represented how many data points that matched a rule belonged to a predicted class.

The most remarkable feature of the set of rules is the very high degree of confidence associated with most rules. Thirteen out of fifteen rules obtain a confidence coefficient of 1.0, which means that all the cases covered by these rules belong to the predicted class. At first sight, this seems to indicate nearly flawless performance. However, when a critical eye is turned upon this situation, it seems that the high confidence levels may pose a problem of possible overtraining, especially for rules which have a very low support. For instance, there are a few rules which assign their classification to a very small set of examples (12 or 24 examples at most) and achieve a confidence coefficient of 1.0.

Rule ID	Cap color	Cap surface	Gill color	Gill size	Habitat	Odor	Population	Ring number	Spore print-color	Stalk root	Stalk shape	Stalk surface	Outcome	Confidence	Support
R1			b, e, g, h				a, c, n, s			b			Poisonous	1	72
R2			b, e, g, h				a, c, n, s			c, e, nan, r			Edible	1	512
R3			b, e, g, h		d, g, l, m, p	a, c, f, l	v, y					f	Poisonous	1	12
R4			b, e, g, h		d, g, l, m, p	m, n, p, s, y	v, y					f	Edible	1	12
R5			b, e, g, h		d, g, l, m, p		v, y					k, s, y	Poisonous	1	2652
R6		f, g	b, e, g, h		u, w		v, y						Edible	1	24
R7		s, y	b, e, g, h		u, w		v, y						Poisonous	1	24
R8			k, n, o, p, r, u, w, y						b, h	b, c			Poisonous	1	624
R9			k, n, o, p, r, u, w, y						b, h	e, nan, r			Edible	1	84
R10			k, n, o, p, r, u, w, y	b				n	k, n, o, r, u, w, y				Poisonous	1	36
R11	b		k, n, o, p, r, u, w, y	b				o, t	k, n, o, r, u, w, y				Edible	0.6	40
R12	c, e, g, n, p, r, u, w, y		k, n, o, p, r, u, w, y	b				o, t	k, n, o, r, u, w, y				Edible	0.9905	3368
R13			k, n, o, p, r, u, w, y	n	d, g				k, n, o, r, u, w, y		e		Poisonous	1	304
R14			k, n, o, p, r, u, w, y	n	l, m, p, u, w				k, n, o, r, u, w, y		e		Poisonous	0.5455	264
R15			k, n, o, p, r, u, w, y	n					k, n, o, r, u, w, y		t		Edible	1	96

Table 3: Mushroom Rulebook Table

Consider Rule R12, which classifies mushrooms as edible with a confidence of 0.9905 and a support of 3,368 observations, making it one of the most influential rules in the model. This rule is based on a combination of gill colour, spore print colour, and ring number, which all were reported as statistically significant during the chi-square analysis. The high support indicates that this rule captures a large and common region of the feature space, while the near perfect confidence suggests a strong and consistent relationship between these observable characteristics and mushroom edibility.

From a strength perspective, Rule R12 epitomises the pragmatic benefit of the rule-based approach. It employs few easily observable features and yields a clear outcome, making the rule well-suited to deployment in a mobile application. The large support is suggestive that the rule is robust and will generalise better than rules built on rare feature combinations.

The existence of two rules with noticeably lower-confidence values (R11, R14) especially demands consideration. These rules identify areas of the feature space in which the overlap of the edible and the poisonous mushrooms become relatively more significant. From a safety point of view, the existence of these ambiguous rules demands especially cautious consideration in how these rules will be implemented, preferably as a trigger to account the mushroom as poisonous.

5. Discussion

This analysis proves that the use of a rule-based-and-data-driven method is efficient in assisting in the identification of edible and poisonous mushrooms based on their attributes. As far as the aim of the National Audubon Society to create a public mobile app is concerned, the identified rules ensure that the method is transparent.

These findings are supported by the consistency of the results with the results of the analysis, as evidenced by the consistency of the analysis results with the behaviour of the models. Features considered to be statistically significant based on chi-square analysis, such as odor, spore print color, and gill attributes, have been confirmed to be important decision points within the rule-set based on their relative ordering.

Limitations:

However, a number of limitations must be noted. Firstly, this dataset is a representation of potential samples of a few species and does not reflect all variability that may be seen foraging in a real-world environment. Observational error and human error may provide a less than truthful representation of feature identification. Additionally, decision tree analysis might place too much importance on strongly related features and fail to accurately generalize rules of lower support.

Overall, this work highlights the value of interpretable, rule-based models in safety-critical contexts while underscoring the importance of cautious deployment and clear communication of uncertainty.

6. Conclusion

In this report, I built a rule-based method to help identify edible and poisonous mushrooms to be used in a mobile app idea that the National Audubon Society has put forward. Through exploratory data analysis, chi-squared analysis, and a simplified decision tree, I derived a set of easily understandable rules that can be applied based on visible characteristics of a mushroom.

These observations emphasized the role of odor, gill type, and spore print color in the identification of mushroom edibility. Although the guidelines are easily understandable and direct, thinking retrospectively reveals that the guidelines are constrained by the dataset and do not promise accurate identification under all scenarios.

I therefore recommend that this rule set be used as a decision-support checklist, not as an identifier, with cautious default assumptions and communication of uncertainties.